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(54) **ADDITIVE MANUFACTURING METHODS
AND SYSTEMS**

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None

See application file for complete search history.

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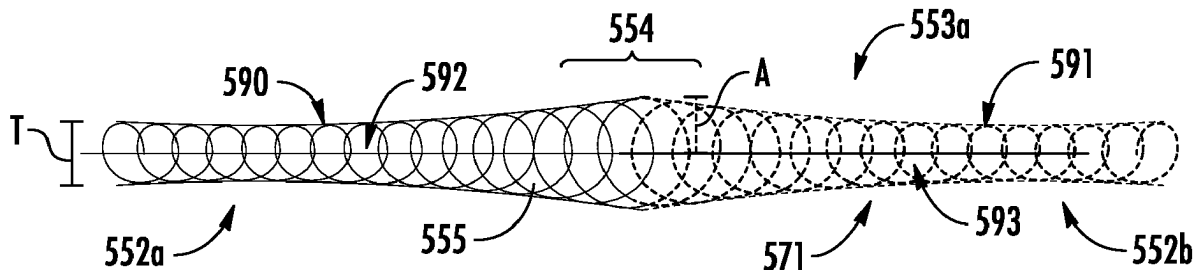
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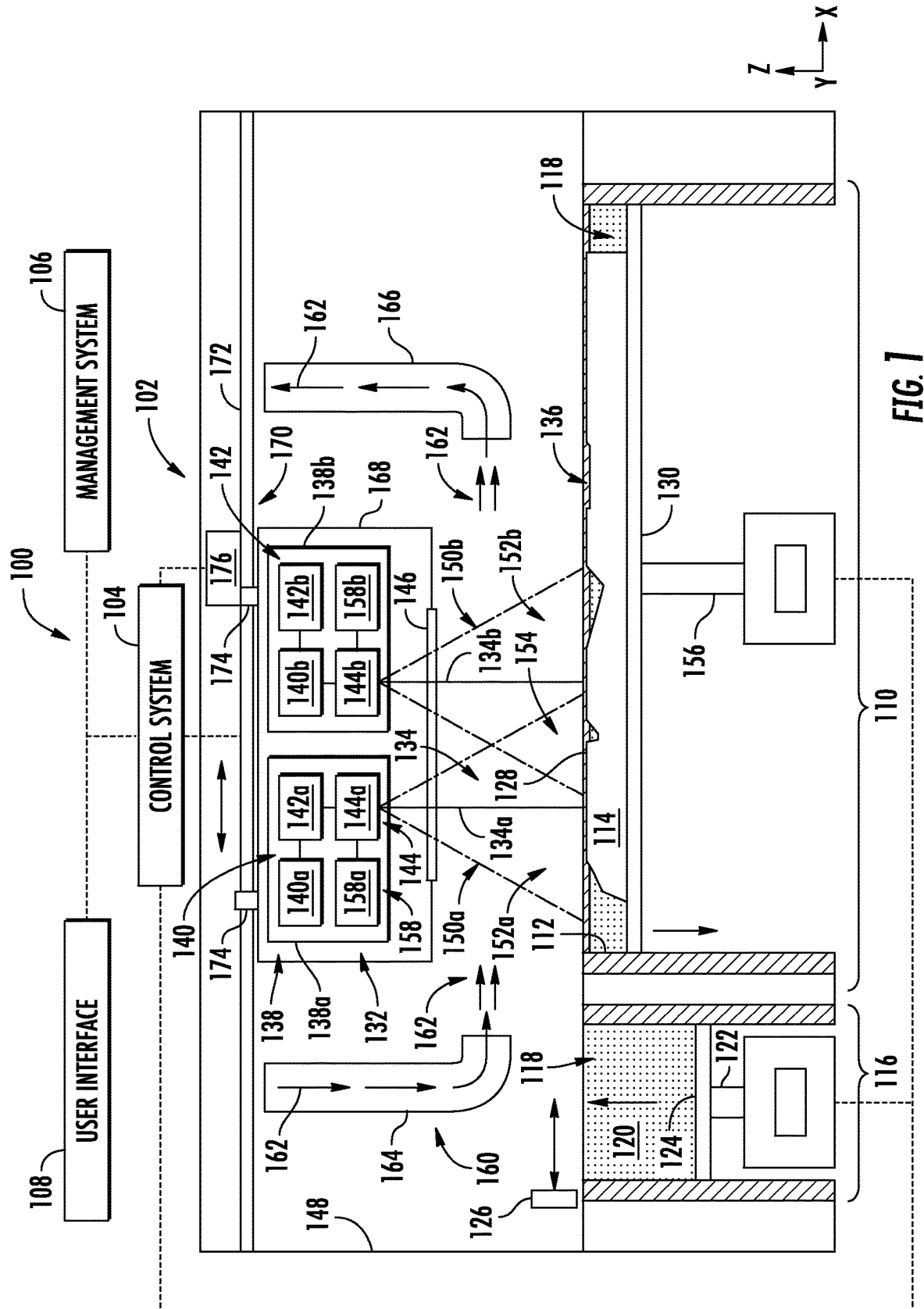
(57) **ABSTRACT**

Methods of additively manufacturing a three-dimensional object include irradiating a first build plane region using a first energy beam defining a beam diameter, the first energy beam travelling along a first oscillating path in a first direction to consolidate a first wall defining a thickness perpendicular to the first direction, wherein a build material adjacent a first side of the first wall and the build material adjacent a second side of the first wall, opposite the first side of the first wall, remains unconsolidated; and wherein the thickness of the first wall is greater than the beam diameter.

16 Claims, 7 Drawing Sheets



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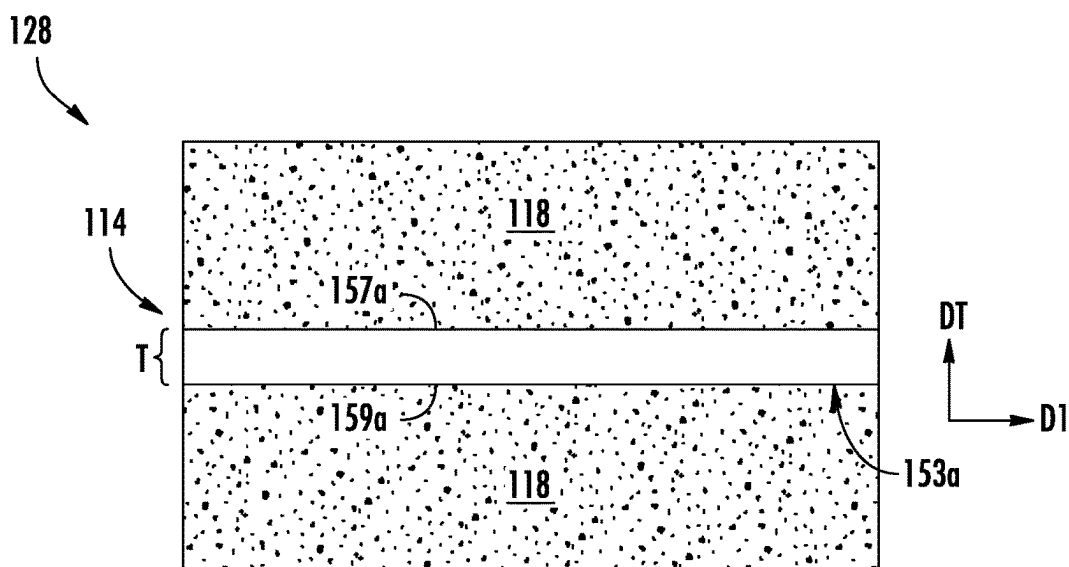


FIG. 2

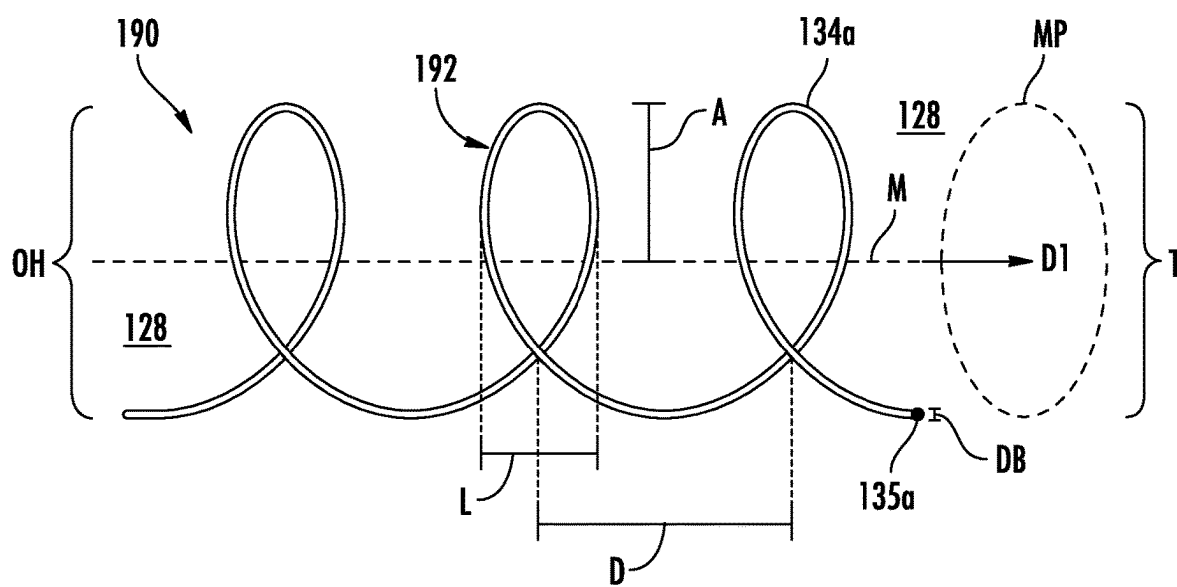


FIG. 3

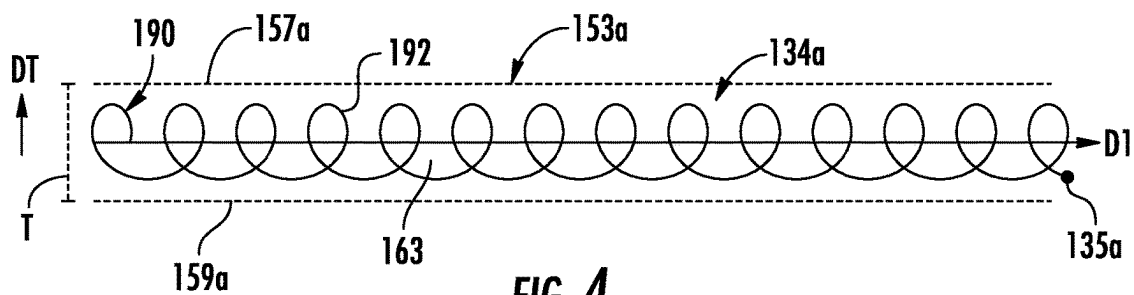


FIG. 4

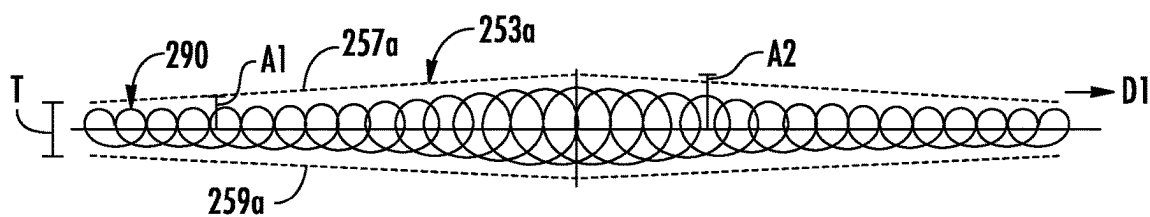


FIG. 5

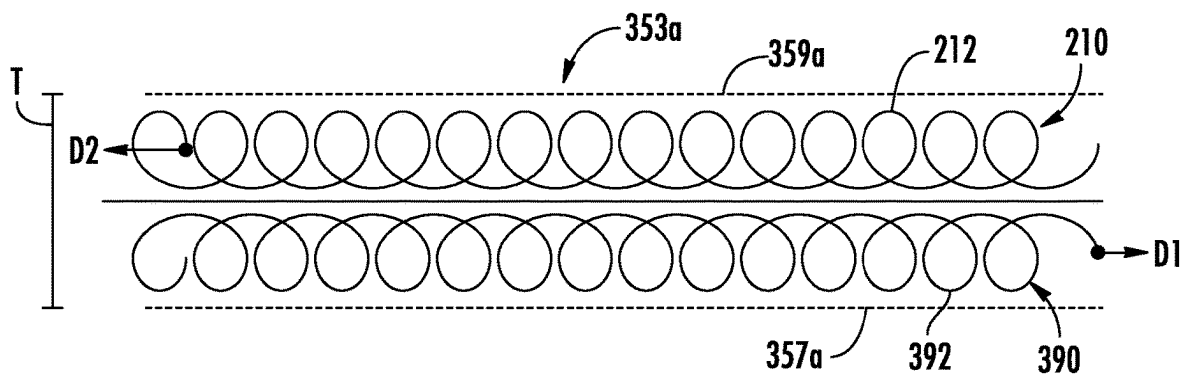
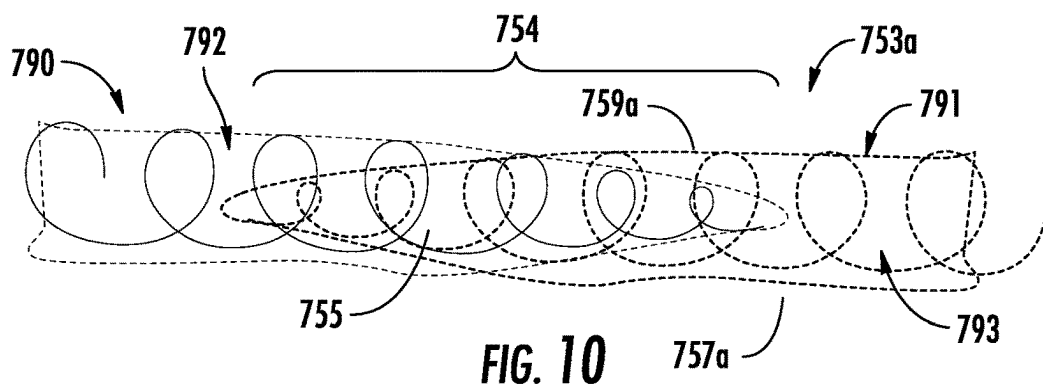
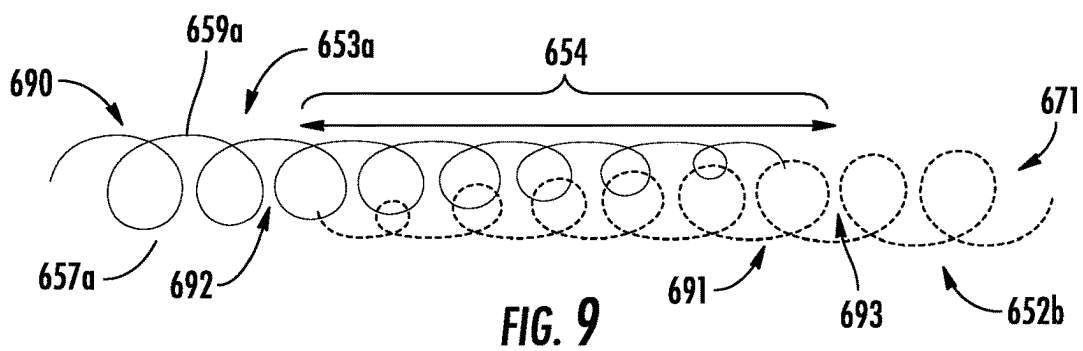
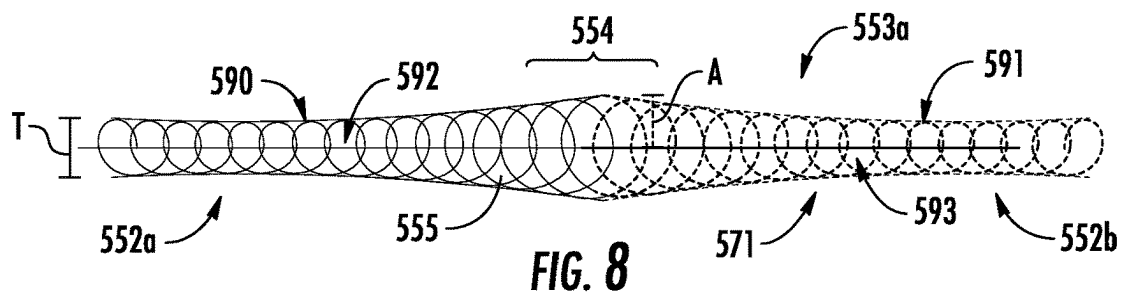
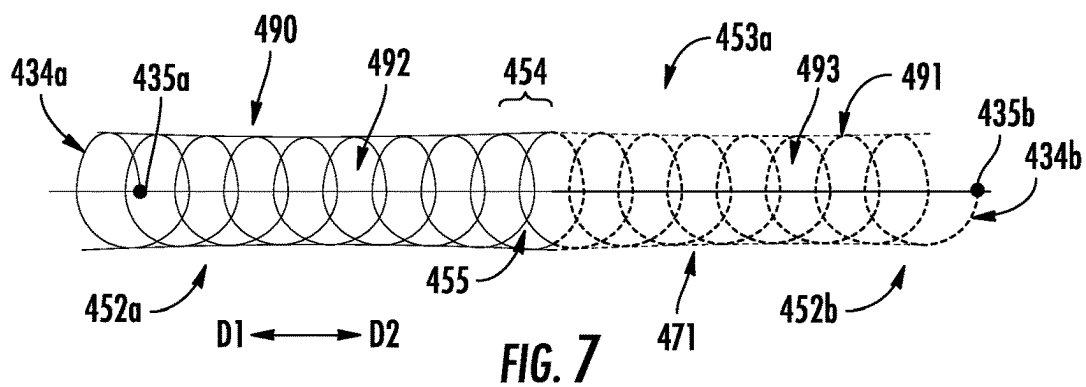
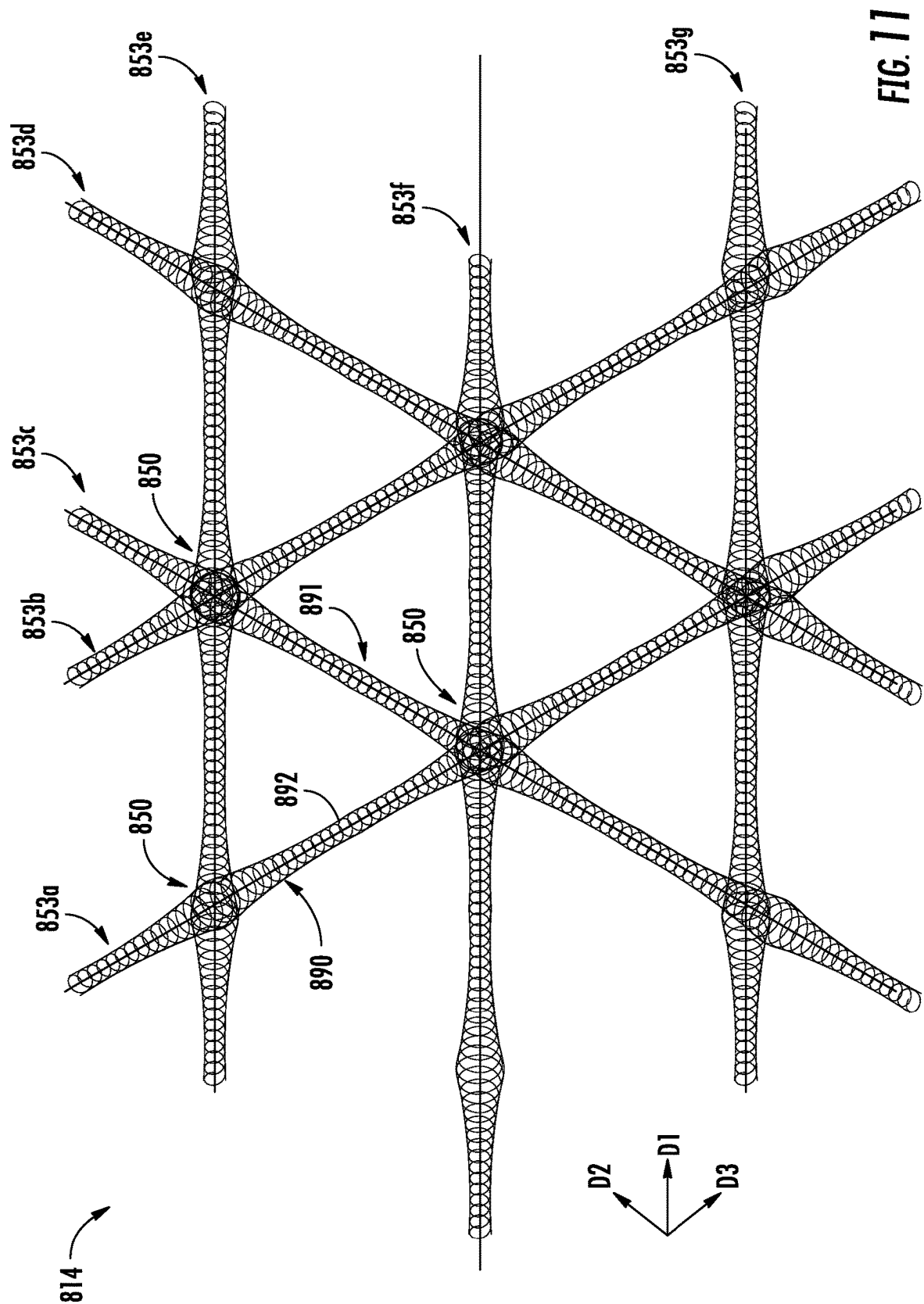
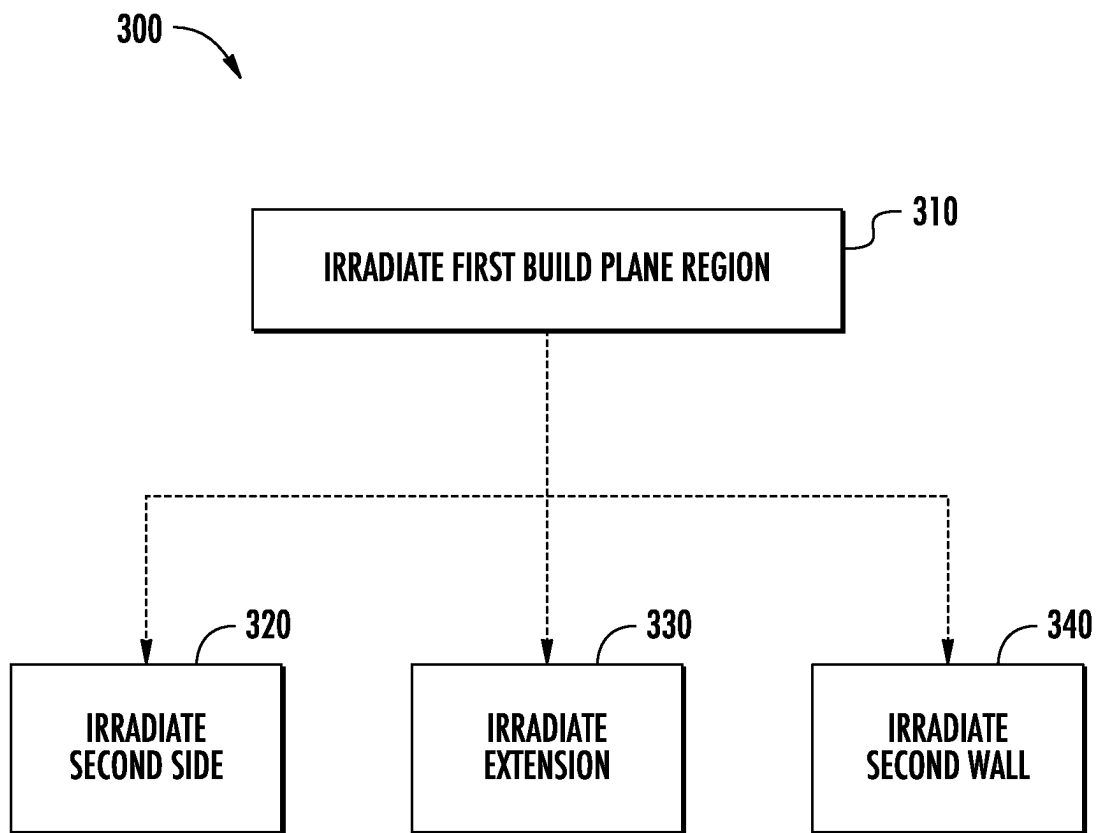


FIG. 6





**FIG. 12**

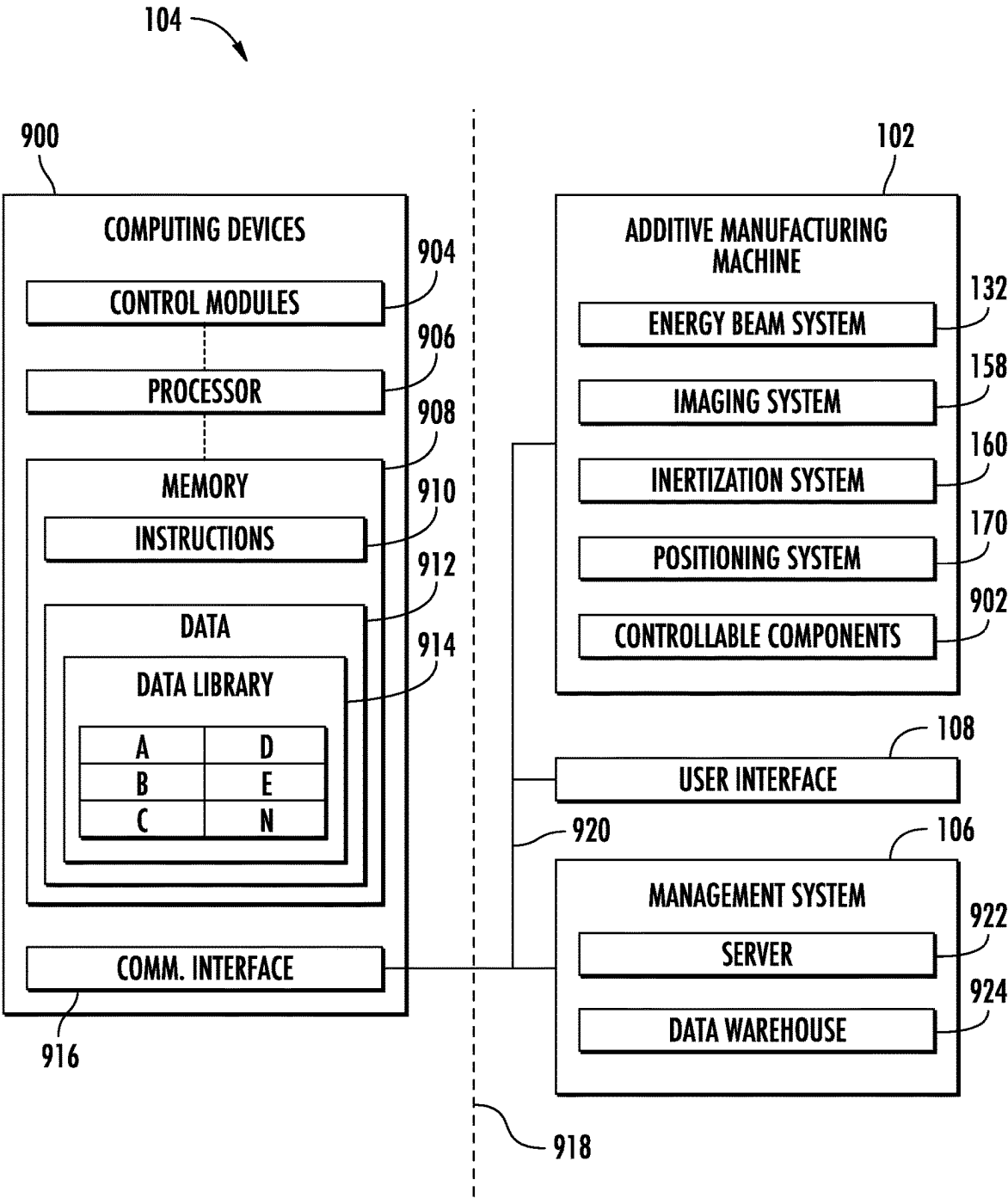


FIG. 13

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ADDITIVE MANUFACTURING METHODS AND SYSTEMS

FIELD

The present disclosure relates to additive manufacturing methods and systems, such as additive manufacturing methods and systems to build thin walls.

BACKGROUND

Three-dimensional objects may be additively manufactured using a variety of methods and systems. For example, additive manufacturing may involve a powder bed fusion process in which one or more energy beams are directed onto a powder bed to melt, fuse, or sinter sequential layers of build material such as powder material. The properties of the three-dimensional object formed by consolidating the powder material depend, at least in part, on one or more parameters of the energy beam. Additionally, one or more parameters of an energy beam impact operating parameters such as processing speed of the additive manufacturing process.

In some additive manufacturing systems, one or more walls may be built using one or more energy beams. For example, a wall is built by consolidating outer edges of the wall with one or more passes by one or more energy beams. The internal portion of the wall can further be consolidated by using additional passes parallel to the edges or by using a distinct hatching pattern to otherwise fill the interior of the wall. The one or more additively manufactured walls can include structures that define, for example, an outer surface or an internal feature of the three-dimensional object, or even provide a lattice support structure to the interior of the three-dimensional object.

BRIEF DESCRIPTION OF THE DRAWINGS

A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

FIG. 1 schematically depicts an exemplary additive manufacturing system in accordance with one or more exemplary aspects of the present disclosure.

FIG. 2 schematically depicts a build plane in accordance with one or more exemplary aspects of the present disclosure.

FIG. 3 schematically depicts an oscillation of an energy beam in accordance with one or more exemplary aspects of the present disclosure.

FIG. 4 schematically depicts an energy beam having a first oscillating path forming a first wall in accordance with one or more exemplary aspects of the present disclosure.

FIG. 5 schematically depicts an energy beam having a first oscillating path with variations forming another first wall in accordance with one or more exemplary aspects of the present disclosure.

FIG. 6 schematically depicts an energy beam having multiple oscillating paths forming yet another first wall in accordance with one or more exemplary aspects of the present disclosure.

FIG. 7 schematically depicts multiple energy beams having multiple oscillating paths to form yet another first wall and an extension in accordance with one or more exemplary aspects of the present disclosure.

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FIG. 8 schematically depicts multiple energy beams having multiple oscillating paths with variations to form yet another first wall and an extension in accordance with one or more exemplary aspects of the present disclosure.

FIG. 9 schematically depicts multiple energy beams having multiple tapered oscillating paths to form yet another first wall and an extension in accordance with one or more exemplary aspects of the present disclosure.

FIG. 10 schematically depicts multiple energy beams having multiple tapered oscillating paths to form yet another first wall and an extension in accordance with one or more exemplary aspects of the present disclosure.

FIG. 11 schematically depicts a three-dimensional object having multiple intersecting walls in accordance with one or more exemplary aspects of the present disclosure.

FIG. 12 schematically depicts an exemplary method of additively manufacturing a three-dimensional object in accordance with one or more exemplary aspects of the present disclosure.

FIG. 13 schematically depicts an exemplary controls system that may be configured to control an additive manufacturing system or machine in accordance with one or more exemplary aspects of the present disclosure.

DETAILED DESCRIPTION

Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

As described herein, the presently disclosed subject matter involves the use of additive manufacturing machines or systems. As used herein, the term “additive manufacturing” refers generally to manufacturing technology in which components are manufactured in a layer-by-layer manner. An exemplary additive manufacturing machine may be configured to utilize any suitable additive manufacturing technology. The additive manufacturing machine may utilize an additive manufacturing technology that includes a powder bed fusion (PBF) technology, such as a direct metal laser melting (DMLM) technology, an electron beam melting (EBM) technology, a selective laser melting (SLM) technology, a directed metal laser sintering (DMLS) technology, or a selective laser sintering (SLS) technology. In an exemplary PBF technology, thin layers of powder material are sequentially applied to a build plane and then selectively melted or fused to one another in a layer-by-layer manner to form one or more three-dimensional objects. Additively manufactured objects are generally monolithic in nature and may have a variety of integral sub-components.

Additionally or alternatively suitable additive manufacturing technologies may include, for example, Fused Deposition Modeling (FDM) technology, Direct Energy Deposition (DED) technology, Laser Engineered Net Shaping (LENS) technology, Laser Net Shape Manufacturing (LNSM) technology, Direct Metal Deposition (DMD) technology, Digital Light Processing (DLP) technology, and other additive manufacturing technologies that utilize an energy beam or other energy source to solidify an additive manufacturing material such as a powder material. In fact, any suitable additive manufacturing modality may be utilized with the presently disclosed the subject matter.

Additive manufacturing technology may generally be described as fabrication of objects by building objects

point-by-point, line-by-line, layer-by-layer, typically in a vertical direction. Other methods of fabrication are contemplated and within the scope of the present disclosure. For example, although the discussion herein refers to the addition of material to form successive layers, the presently disclosed subject matter may be practiced with any additive manufacturing technology or other manufacturing technology, including layer-additive processes, layer-subtractive processes, or hybrid processes.

The additive manufacturing processes described herein may be used for forming components using any suitable material. For example, the material may be metal, ceramic, polymer, epoxy, photopolymer resin, plastic, or any other suitable material that may be in solid, powder, sheet material, wire, or any other suitable form, or combinations thereof. Additionally, or in the alternative, exemplary materials may include metals, ceramics, or binders, as well as combinations thereof. Exemplary ceramics may include ultra-high-temperature ceramics, or precursors for ultra-high-temperature ceramics, such as polymeric precursors. Each successive layer may be, for example, between about 10 μm and 200 μm , although the thickness may be determined based on any number of parameters and may be any suitable size. Moreover, the additive manufacturing process described herein may be used for forming any type of suitable component. For example, the component formed using the additive manufacturing process described herein may comprise one or more turbine components such as turbine blades, shrouds, nozzles, heat shields, or vanes.

As used herein, the term “build plane” refers to a plane defined by a surface upon which an energy beam impinges to selectively irradiate and thereby consolidate powder material during an additive manufacturing process. Generally, the surface of a powder bed defines the build plane. During irradiation of a respective layer of the powder bed, a previously irradiated portion of the respective layer may define a portion of the build plane. Prior to distributing powder material across a build module, a build plate that supports the powder bed generally defines the build plane.

As used herein, the term “consolidate” or “consolidating” refers to solidification of build material (e.g., powder material) as a result of irradiating the build material, including by way of melting, fusing, sintering, or the like such that multiple separate pieces of build material (e.g., multiple individual pieces of powder material) join together into a single structure.

As used herein, the term “unconsolidated” refers to separate pieces of material that are not bonded or otherwise joined to one another, such as separate pieces of loose powder.

It is also understood that terms such as “top”, “bottom”, “outward”, “inward”, and the like are words of convenience and are not to be construed as limiting terms.

As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

The terms “a” and “an” do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item.

Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “substantially,” and “approximately,” are not to be limited to the precise value specified. In at least

some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components or systems. For example, the approximating language may refer to being within a 10 percent margin.

Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

The present disclosure is generally related to additive manufacturing methods and systems to build thin walls for three-dimensional objects. One or more energy beams in such systems can be used to build walls to define, for example, an outer surface, an internal feature, or an internal support lattice of a three-dimensional object. However, the minimal obtainable thickness of the wall may be inhibited by requiring multiple passes of an energy beam. Moreover, the uniformity of the consolidated build material within the wall may be inhibited when utilizing multiple passes or multiple energy beams.

Accordingly, alternative additive manufacturing methods and systems would be welcomed in the art, including additive manufacturing methods and systems that provide thin walls using energy beam oscillations.

The presently disclosed subject matter will now be described in further detail. FIG. 1 schematically depicts an additive manufacturing system 100. The additive manufacturing system 100 may include one or more additive manufacturing machines 102. It will be appreciated that the additive manufacturing system 100 and additive manufacturing machine 102 shown in FIG. 1 is provided by way of example and not to be limiting. In fact, the subject matter of the present disclosure may be practiced with any additive manufacturing system 100 and additive manufacturing machine 102 without departing from the scope of the present disclosure. As shown, the one or more additive manufacturing machines 102 may include a control system 104. The control system 104 may be included as part of the additive manufacturing machine 102 or the control system 104 may be associated with the additive manufacturing machine 102. The control system 104 may include componentry integrated as part of the additive manufacturing machine 102 or componentry that is provided separately from the additive manufacturing machine 102. Various componentry of the control system 104 may be communicatively coupled to various componentry of the additive manufacturing machine 102.

The control system 104 may be communicatively coupled with a management system 106 or a user interface 108. The management system 106 may be configured to interact with the control system 104 in connection with enterprise-level operations pertaining to the additive manufacturing system 100. Such enterprise level operations may include transmitting data from the management system 106 to the control system 104 or transmitting data from the control system 104 to the management system 106. The user interface 108 may include one or more user input/output devices to allow a user to interact with the additive manufacturing system 100.

As shown, an additive manufacturing machine 102 may include a build module 110 that includes a build chamber 112 within which an object or three-dimensional objects 114 may be additively manufactured. An additive manufacturing machine 102 may include a powder module 116 that con-

tains a supply of build material **118** (e.g., powder material) housed within a supply chamber **120**. The build module **110** or the powder module **116** may be provided in the form of modular containers configured to be installed into and removed from the additive manufacturing machine **102** such as in an assembly-line process. Additionally, or in the alternative, the build module **110** or the powder module **116** may define a fixed componentry of the additive manufacturing machine **102**.

The powder module **116** contains a supply of build material **118** housed within a supply chamber **120**. The powder module **116** includes a powder piston **122** that elevates a powder floor **124** during operation of the additive manufacturing machine **102**. As the powder floor **124** elevates, a portion of the build material **118** is forced out of the powder module **116**. A recoater **126** such as a blade or roller sequentially distributes thin layers of build material **118** across a build plane **128** above the build module **110**. A build platform **130** supports the sequential layers of build material **118** distributed across the build plane **128**. A build platform **130** may include a build plate (not shown) secured thereto and upon which a three-dimensional object **114** may be additively manufactured.

The additive manufacturing machine **102** includes an energy beam system **132** configured to generate one or more energy beams **134** and to direct the energy beams **134** onto the build plane **128** to selectively solidify respective portions of the powder bed **136** defining the build plane **128**. The energy beams **134** may be laser beams or beams from any other suitable energy source, such as LEDs or other light sources, and so forth. As the energy beams **134** selectively melt or fuse the sequential layers of build material **118** that define the powder bed **136**, the three-dimensional object **114** begins to take shape. The one or more energy beams **134** or laser beams may include electromagnetic radiation having any suitable wavelength or wavelength range, such as a wavelength or wavelength range corresponding to infrared light, visible light, or ultraviolet light, or a combination thereof.

Typically, with a DMLM, EBM, or SLM system, the build material **118** is fully melted, with respective layers being melted or re-melted with respective passes of the energy beams **134**. With DMLS or SLS systems, typically the layers of build material **118** are sintered, fusing particles of build material **118** to one another generally without reaching the melting point of the build material **118**. The energy beam system **132** may include componentry integrated as part of the additive manufacturing machine **102** or componentry that is provided separately from the additive manufacturing machine **102**.

The energy beam system **132** may include one or more irradiation devices **138** configured to generate a plurality of energy beams **134** and to direct the energy beams upon the build plane **128**. An energy beam system **132** may include a plurality of irradiation devices **138**, such as a first irradiation device **138a** and a second irradiation device **138b**. The one or more irradiation devices **138** may respectively include an energy beam source **140** (e.g., first energy beam source **140a** and second energy beam source **140b**), an optical assembly **142** (e.g., first optical assembly **142a** and second optical assembly **142b**), and a scanner **144** (e.g., first scanner **144a** and second scanner **144b**). The optical assembly **142** may include a plurality of optical elements configured to direct the energy beam onto the build plane **128**. The optical assembly **142** may include one or more optical elements, such as lenses through which an energy beam may be transmitted along an optical path from the energy beam

source to the build plane. By way of example, an optical assembly **142** may include one more focusing lenses that focus an energy beam **134** on a build plane **128**. A scanner **144** may include a galvo scanner, an electro-optic modulator, an acousto-optic modulator, a piezo-driven mirror, or the like. Additionally, or in the alternative, the energy beam system **132** may include a window **146**, such as a protective glass, that separates one or more components of the energy beam system **132** from the environment of a process chamber **148** within which build material **118** is irradiated by the one or more energy beams **134** to additively manufacture a three-dimensional object **114**.

The window **146** may prevent contaminants from fumes associated with the additive manufacturing process, such as powder material, dust, soot, residues, vapor, byproducts, and the like, from coming into contact with sensitive components of an energy beam system **132**. Accumulation of contaminants upon various optical elements of an optical assembly **142** may adversely affect operation of the energy beam system **132** or quality metrics associated with an energy beam system. Additionally, or in the alternative, such contaminants may cause damage to various optical elements of an optical assembly **142**.

As shown in FIG. 1, the energy beam system **132** includes a first irradiation device **138a** and a second irradiation device **138b**. Additionally, or in the alternative, an energy beam system **132** may include any number of additional irradiation devices such as three, four, six, eight, ten, or more irradiation devices, and such irradiation devices may respectively include an optical assembly **142**. The plurality of irradiation devices **138** may be configured to respectively generate one or more energy beams that are respectively scannable within a scan field incident upon at least a portion of the build plane **128** to selectively consolidate the portions of the build material **118** that are to become part of the three-dimensional object **114**.

For example, the first irradiation device **138a** may generate a first energy beam **134a** that is scannable within a first scan field **150a** incident upon at least a first build plane region **152a**. The second irradiation device **138b** may generate a second energy beam **134b** that is scannable within a second scan field **150b** incident upon at least a second build plane region **152b**. The first scan field **150a** and the second scan field **150b** may overlap such that the first build plane region **152a** scannable by the first energy beam **134a** overlaps with the second build plane region **152b** scannable by the second energy beam **134b**. The overlapping portion of the first build plane region **152a** and the second build plane region **152b** may sometimes be referred to as an interlace region **154**. Portions of the powder bed **136** to be irradiated within the interlace region **154** may be irradiated by the first energy beam **134a**, or the second energy beam **134b**, or a combination thereof. While the powder bed **136** to be irradiated is exemplary illustrated as being irradiated by the first energy beam **134a** or the second energy beam **134b**, it is appreciated that any number of energy beams **134** may additionally or alternatively be utilized for irradiating the build plane **128** in accordance with the present disclosure.

To irradiate a layer of the powder bed **136**, the one or more irradiation devices **138** (e.g., the first irradiation device **138a** and the second irradiation device **138b**) respectively direct the plurality of energy beams **134** beams (e.g., the first energy beam **134a** and the second energy beam **134b**) across the respective portions of the build plane **128** (e.g., the first build plane region **152a** and the second build plane region **152b**) to selectively consolidate the portions of the build material **118** that are to become part of the three-dimensional

object **114**. The one or more energy beams **134** may become incident upon the build plane **128** defined by the powder bed **136**, for example, after passing through one or more optical elements of the optical assembly **142** or through a window **146** of the energy beam system **132**. As sequential layers of the powder bed **136** are consolidated, a build piston **156** gradually lowers the build platform **130** to make room for sequential layers of build material **118**. As sequential layers of build material **118** are applied across the build plane **128**, the next sequential layer of build material **118** defines the surface of the powder bed **136** coinciding with the build plane **128**. Sequential layers of the powder bed **136** may be selectively consolidated until a completed object **114** has been additively manufactured. In some aspects of the disclosure, an additive manufacturing machine may utilize an overflow module (not shown) to capture excess build material **118**. Additionally, or in the alternative, excess build material **118** may be redistributed across the build plane **128** when applying a next sequential layer of build material **118**. It will be appreciated that other systems may be provided for handling the build material **118**, including different powder supply systems or excess powder recapture systems. The subject matter of the present disclosure may be practiced with any suitable additive manufacturing machine without departing from the scope hereof.

Still referring to FIG. 1, an additive manufacturing machine **102** may include an imaging system **158** (e.g., first imaging system **158a** and second imaging system **158b**) configured to monitor one or more operating parameters of an additive manufacturing machine **102**, one or more parameters of an energy beam system **132**, or one or more operating parameters of an additive manufacturing process. The imaging system may have a calibration system configured to calibrate one or more operating parameters of an additive manufacturing machine **102** or of an additive manufacturing process. The imaging system **158** may be a melt pool monitoring system. The one or more operating parameters of the additive manufacturing process may include operating parameters associated with additively manufacturing a three-dimensional object **114**. The imaging system **158** may be configured to detect an imaging beam such as an infrared beam from a laser diode or a reflected portion of an energy beam (e.g., a first energy beam **134a** or a second energy beam **134b**).

An energy beam system **132** or an imaging system **158** may include one or more detection devices. The one or more detection devices may be configured to determine one or more parameters of an energy beam system **132**, such as one or more parameters associated with irradiating the sequential layers of the powder bed **136** based at least in part on an assessment beam detected by the imaging system **158**. One or more parameters associated with consolidating the sequential layers of the powder bed **136** may include irradiation parameters or object parameters, such as melt pool monitoring parameters. The one or more parameters determined by the imaging system **158** may be utilized, for example, by the control system **104**, to control one or more operations of the additive manufacturing machine **102** or of the additive manufacturing system **100**. The one or more detection devices may be configured to obtain assessment data of the build plane **128** from a respective assessment beam. An exemplary detection device may include a camera, an image sensor, a photo diode assembly, or the like. For example, a detection device may include charge-coupled device (e.g., a CCD sensor), an active-pixel sensor (e.g., a complementary metal-oxide semiconductor (CMOS) sensor), a quanta image device (e.g., a QIS sensor), or the like.

A detection device may additionally include a lens assembly configured to focus an assessment beam along a beam path to the detection device. An imaging system **158** may include one or more imaging optical elements (not shown), such as mirrors, beam splitters, lenses, and the like, configured to direct an assessment beam to a corresponding detection device.

In addition, or in the alternative, to determine parameters associated with irradiation the sequential layers of the powder bed **136**, the imaging system **158** may be configured to perform one or more calibration operations associated with an additive manufacturing machine **102**, such as a calibration operation associated with the energy beam system **132**, one or more irradiation devices **138** or components thereof, or the imaging system **158** or components thereof. The imaging system **158** may be configured to project an assessment beam and to detect a portion of the assessment beam reflected from the build plane **128**. The assessment beam may be projected by an irradiation device **138** or a separate beam source associated with the imaging system **158**. Additionally, or in the alternative, the imaging system **158** may be configured to detect an assessment beam that includes radiation emitted from the build plane **128**, such as radiation from an energy beam **134** reflected from the powder bed **136** or radiation emitted from a melt pool in the powder bed **136** generated by an energy beam **134** or radiation emitted from a portion of the powder bed **136** adjacent to the melt pool. The imaging system **158** may include componentry integrated as part of the additive manufacturing machine **102** or componentry that is provided separately from the additive manufacturing machine **102**. For example, the imaging system **158** may include componentry integrated as part of the energy beam system **132**. Additionally, or in the alternative, the imaging system **158** may include separate componentry, such as in the form of an assembly, that can be installed as part of the energy beam system **132** or as part of the additive manufacturing machine **102**.

Still referring to FIG. 1, in some aspects of the disclosure, an inertization system **160** may supply a flow of inert process gas **162** to one or more regions the process chamber **148**, such as a region between the energy beam system **132** and the powder bed **136**. The flow of inert process gas **162** may remove fumes from the process chamber **148** or to reduce the tendency of fumes to interfere with the energy beams **134** used to irradiate the build material **118**. Such fumes may present in the form of a plume emanating from a consolidation zone where an energy beam **134** becomes incident upon the powder bed **136** and may sometimes be referred to as a fume plume. A fume plume may include build material, dust, soot, residues, vapors, byproducts, or the like. The flow of inert process gas **162** may also reduce the tendency of contaminants from fumes to deposit on the window **146**, optical elements of the optical assembly **142**, or other components of the energy beam system **132**. The inertization system **160** may provide a directional flow of inert process gas **162** that flows across the build plane **128**. For example, as shown, the inert process gas **162** flows from left to right. The inertization system **160** may include a supply manifold **164** and a return manifold **166**. The inert process gas **162** may flow from the supply manifold **164** to the return manifold **166**. Fumes in the process chamber **148** may be drawn into the return manifold **166**. In some aspects of the disclosure, the supply manifold **164** or the return manifold **166** may be coupled to, or define a portion of, a perimeter wall of the process chamber **148**. Additionally, or in the alternative, the supply manifold **164** or the return manifold **166** may be coupled to a housing assembly **168**.

that contains one or more components of the energy beam system 132, such as one or more irradiation devices 138 and or one or more imaging systems 158. With the supply manifold 164 or the return manifold 166 coupled to the housing assembly 168, a relatively small volume of space between the energy beam system 132 and the powder bed 136 may be inertized, as opposed to inertizing an entire interior of the process chamber 148. Additionally, or in the alternative, a fume plume may have a shorter path to travel before being drawn into the return manifold 166 by the flow of inert process gas 162.

The energy beam system 132 may be positioned at any suitable location within the process chamber 148. Additionally, or in the alternative, the energy beam system 132 may be coupled to a perimeter wall of the process chamber 148. In some aspects of the disclosure, an additive manufacturing machine may include a positioning system 170 configured to move an energy beam system 132 or one or more components thereof relative to the build plane 128. The positioning system 170 may be configured to move the energy beam system 132 or one or more components thereof to specified build coordinates or along specified build vectors corresponding to a cartesian coordinate system in accordance with control commands provided, for example, by the control system 104. The control commands may be provided, for example, to carry out operations of the one or more energy beam system 132 or of the additive manufacturing machine 102 in accordance with the present disclosure. The positioning system 170 may include one or more gantry elements 172 configured to move the energy beam system 132 or one or more components thereof across the powder bed. The gantry elements 172 may respectively be configured to move the energy beam system 132 or one or more components thereof in one or more directions, such as an X-direction, a Y-direction, or a Z-direction. In some aspects of the disclosure, the positioning system 170 may be coupled to the housing assembly 168 that contains one or more components of the energy beam system 132. The housing assembly 168 may be coupled to one or more gantry elements 172 by one or more gantry mounts 174. The positioning system 170 may include a drive motor 176 configured to move the housing assembly 168 or the one or more components the energy beam system 132 according to instructions for the control system 104. The positioning system 170 may include componentry typically associated with a gantry system, such as stepper motors, drive elements, carriages, and so forth.

Referring now additionally to FIG. 2, a top down view of a build plane 128 is illustrated for forming a first wall 153a of the three-dimensional object 114. The first wall 153a generally refer to any thin or narrow structure consolidated by one or more energy beams 134 (not shown in FIG. 2). The first wall 153a extends in a first direction D1 and can be defined by a first side 157a of the first wall 153a and a second side 159a of the first wall 153a, wherein the second side 159a is opposite the first side 157a. Build material 118 adjacent the first side 157a of the first wall 153a and adjacent the second side 159a of the first wall 153a remains unconsolidated. That is, the build material 118 adjacent the first wall 153a remains in its form (e.g., loose powder material) when applying the build material 118 to the build plane 128.

The first wall 153a may comprise a variety of configurations and orientations. For example, the first wall 153a may comprise a generally linear orientation such as illustrated in FIG. 2. In some aspects of the disclosure, the first wall 153a may comprise a non-linear orientation such that it

includes one or more bends, turns, curves or the like. Further, the first wall 153a comprises a thickness T in the thickness direction DT which is perpendicular to the first direction D1 (that is, the distance between the first side 157a of the first wall 153a and the second side 159a of the first wall 153a). In some aspects of the disclosure, the first wall 153a may comprise a substantially uniform thickness T as it extends in the first direction D1. However, in some aspects of the disclosure, the thickness T of the first wall 153a may increase or decrease at one or more locations along the first direction D1.

Further, while FIG. 2 illustrates a first wall 153a in isolation within the build plane 128, it is appreciated that other structures may also be consolidated in the build plane 128 at one or more locations. For example, the first wall 153a may extend between and connect to one or more larger structures separated by a defined distance. Alternatively, or additionally, the first wall 153a may intersect with additional structures in the build plane 128.

Referring now additionally to FIG. 3, a top down view of a first oscillating path 190 is illustrated for consolidating the first wall 153a illustrated in FIG. 2. The first oscillating path 190 generally comprises a plurality of oscillations 192 that repeat along a first direction D1. That is, the first spot 135a of the first energy beam 134a can wobble as it travels across the build plane 128 in the first direction D1 to produce the plurality of oscillations 192. The wobbling of the first energy beam 134a can allow for a larger melt pool in the build plane 128 than if the first spot 135a were to merely travel linearly in parallel with the first direction D1. For instance, the first spot 135a comprises a beam diameter DB defining the diameter of the first spot 135a on the build plane 128. The melt pool MP formed in the build plane 128 by the plurality of oscillations 192 can exceed the size of the beam diameter DB to consolidate a larger area via a single pass or fewer passes. For example, due in part to the plurality of oscillations 192, the thickness T of the first wall 153a can be greater than the beam diameter DB. In some aspects of the disclosure, the thickness T of the first wall 153a may be between 1.5 times and seven times the beam diameter DB of the first energy beam 134a. In some aspects of the disclosure, the thickness T of the first wall 153a may be between two times and five times the beam diameter DB of the first energy beam 134a. In some aspects of the disclosure, the thickness T of the first wall 153a may be between two times and three times the beam diameter DB of the first energy beam 134a.

The first oscillating path 190 can comprise a variety of parameters and configurations including with respect to the plurality of oscillations 192. For instance, each oscillation 192 may generally comprise an amplitude A and a length L. The amplitude A refers to the maximum distance reached by the first spot 135a extending away from a midline M which bisects the first oscillating path 190 in the first direction D1D1. The length L refers to the distance extending in the first direction D1D1 of each oscillation 192. Moreover, the first oscillating path 190 can comprise a distance D between oscillations 192.

The amplitude A of each oscillation 192 may impact the size of the overall melt pool, which can have a diameter the same as or similar to the overall oscillation height OH. For instance, a greater amplitude can create a greater melt pool size which will, in turn, lead to the consolidation of larger amount of build material 118 (FIGS. 1 and 2) in the build plane 128 from a single pass of the first energy beam 134a. The length L of each oscillation 192 may impact the amount of energy imparted on the build material 118 (FIGS. 1 and

2) in the build plane 128 from the first energy beam 134a by adjusting the amount of distance travelled by the first spot 135a of the first energy beam 134a over any particular area. A smaller length L can produce a greater travel distance of the first spot 135a over a smaller area along the direction of travel to increase the overall amount of energy imparted on the build material 118 (FIGS. 1 and 2). Further, the frequency of oscillations 192 in the first oscillating path 190 can be tailored by adjusting the distance D between oscillations 192. A smaller distance D results in oscillations that are more frequent which in turn, similar to a smaller length L, can produce a greater travel distance of the first spot 135a over a smaller area along the travel direction D1 to increase the overall amount of energy imparted on the build material 118 (FIGS. 1 and 2). The length L and distance D parameters may be adjusted to any suitable values that produce a suitable melt pool in the build plane 128. In some aspects of the disclosure, such as that illustrated in FIG. 3, the oscillations 192 may be separated from one another. However, in some aspects of the disclosure, the length L may be large enough or the distance D may be small enough that the oscillations 192 may partially overlap with one another along the travel direction T.

In some aspects of the disclosure, the amplitude A, length L, and distance D for each oscillation 192 can each remain constant along the first oscillating path 190. However, in some aspects of the disclosure, one or more of the amplitude A, length L, and distance D can independently vary along the first oscillating path 190 within oscillations 192. For example, the amplitude A may increase for certain oscillations over a certain distance to temporarily grow the size of the melt pool with respect to the first oscillating path 190.

The oscillations 192 may comprise a variety of configurations. In some aspects of the disclosure, such as that illustrated in FIG. 3, the oscillations 192 may comprise symmetrical loops. In some aspects of the disclosure, one or more of the oscillations 192 may additionally or alternatively comprise other configurations, such as asymmetrical loops, linear or non-linear patterns, or combinations thereof. Moreover, while the first direction D1 is illustrated as comprising a relatively linear path, it is appreciated that the first direction D1 may additionally, or alternatively, comprise one or more non-linear portions such as bends, curves, turns, or the like.

Referring now additionally to FIG. 4, the plurality of oscillations 192 of the first oscillating path 190 can irradiate the build material 118 (FIGS. 1 and 2) to consolidate at least a portion of the first wall 153a. Moreover, the plurality of oscillations 192 may define at least the first side 157a of the first wall 153a by irradiating and consolidating the build material 118 at the edge of the first wall 153a without the need for additional irradiation or passes from energy beams 134 (not shown). The irradiation imparted by the first spot 135a of the first energy beam 134a at the peak amplitude A of each oscillation 192 will thereby cause the first side 157a and the second side 159a of the first wall 153a to be consolidated to define the same. That is, the amplitude A will control the maximum dimension of the melt pool, which, in turn, becomes the outer surface of the first side 157a and the second side 159a of the first wall 153a. Thus, a constant amplitude A will produce a linear first side 157a and second side 159a of the first wall 153a. However, a varying amplitude A will produce a wall 153a having a first side 157a or second side 159a that is nonlinear such as expanding and retracting in the direction of the thickness direction DT. Such aspects of the disclosure can allow for relatively thin-walled structures to be built for three-dimensional

objects 114 with higher efficiency, precision, and uniformity compared to additively manufacturing walls using non-oscillating laser paths.

The body portion 163 of the first wall 153a, i.e., the portion between the first side 157a of the first wall 153a and the second side 159a of the first wall 153a, can be consolidated using the rest of the oscillating path 190. Such aspects of the disclosure can avoid the need for hatching (e.g., a plurality of linear passes by one or more energy beams 134) to help speed up the manufacturing process and avoid potential solidification inconsistencies that could occur from multiple adjacent passes.

In some aspects of the disclosure, the plurality of oscillations 192 may further define the second side 159a of the first wall 153a by irradiating and consolidating the build material 118 at the other edge of the first wall 153a without the need for additional irradiation or passes from energy beams 134. The irradiation imparted by the first spot 135a of the first energy beam 134a at the peak amplitude A of each oscillation 192 will thereby also cause the second side 159a of the first wall 153a to be consolidated to define the same. Such aspects of the disclosure can allow for a single pass of the first energy beam 134a on the first oscillating path 190 to define the entirety of the first wall 153a.

FIG. 5 illustrates another first oscillating path 290 according to another aspect of the disclosure. The first oscillating path 290 is similar to first oscillating path 190; therefore, like components will be identified with like numerals increased by 100, with it being understood that the description of the like parts of the first oscillating path 190 applies to the first oscillating path 290, unless otherwise noted. As illustrated in FIG. 5, one or more dimensions of the first oscillating path 290 are varied as the first energy beam 134a (FIG. 1) travels in the first direction D1. For example, as illustrated in FIG. 5, the amplitude A of the first oscillating path 290 may vary as the first oscillating path travels in the first direction D1. In such aspects of the disclosure, the first oscillating path may comprise at least a first amplitude A1 at a first location and a second amplitude A2 at a second location further along in the first direction D1. The second amplitude A2 can be greater than the first amplitude A1 such that a thickness T of the first wall 253a between the first side 257a and the second side 259a can increase along a first portion of the first wall 253a. Depending on the design of the first wall 253a and the three-dimensional object 114, the amplitude A can vary (e.g., increase or decrease) at a single location or at a plurality of locations. Moreover, the amplitude A may steadily vary (e.g., a tapered increase or tapered decrease) or may quickly vary (e.g., an immediate increase or immediate decrease), or a combination thereof as the first oscillating path 190 extends in the first direction D1. Further, a power of the first energy beam can be adjusted when the plurality of oscillations is varied, such as by increasing or decreasing the power when the amplitude A is increased or decreased.

The first energy beam 134a (FIG. 1) may comprise a variety of beam parameters along the first oscillating path such as, but not limited to, power, spot size, focus depth, travel speed, and other beam parameters for sufficient consolidation of build material 118. Similar to the amplitude A, the beam parameters of the first energy beam 134a may remain constant throughout the entire oscillating path 190, or may vary in whole or in part throughout the oscillating path 190.

FIG. 6 illustrates another first oscillating path 390 according to another aspect of the disclosure. The first oscillating path 390 is similar to first oscillating path 190; therefore,

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like components will be identified with like numerals increased by 100, with it being understood that the description of the like parts of the first oscillating path 190 applies to the first oscillating path 290, unless otherwise noted. As illustrated in FIG. 6, the first wall 353a may be consolidated using additional irradiation such as additional passes by the first energy beam 134a (FIG. 1) or additional passes by the second energy beam 134b (FIG. 1). For example, with reference to FIG. 6, in some aspects of the disclosure, a second build plane region 152b (FIG. 1) may be irradiated along a second oscillating path 210 in a second direction D2 that is opposite the first direction D1 to consolidate the first wall 353a. The second oscillating path 210 can comprise a plurality of oscillations 212 that define the second side 359a of the first wall 353a, which is opposite the first side 357a of the first wall 352a to collectively define the thickness T of the first wall 352a. As such, the first oscillating path 390 and the second oscillating path 210 may traverse adjacent and substantially parallel to one another to consolidate the first wall 353a. While exemplary illustration is made to the first energy beam 134a and the second energy beam 134b, it is appreciated that any number of energy beams 134 may additionally or alternatively be utilized for irradiating the build plane 128 in any number of regions.

In some aspects of the disclosure, the first energy beam 134a may travel both the first oscillating path 390 and the second oscillating path 210. Alternatively, in some aspects of the disclosure, the first energy beam 134a may travel the first oscillating path 390 and the second energy beam 134b may travel the second oscillating path 210. Moreover, the first oscillating path 390 and the second oscillating path 210 may generally comprise the same dimensions such as the same amplitude A, length L, or distance D (FIG. 3). In some aspects of the disclosure, one or more of the dimensions may vary between the first oscillating path 390 and the second oscillating path 210. Furthermore, the parameters of the energy beam 134 used for the first oscillating path 390 and the second oscillating path 210 may be similar to one another, dissimilar to one another, or a combination thereof.

Referring now to FIGS. 7-10, in some aspects of the disclosure, different oscillating paths may form adjacent portions parts of the same wall, wherein the different oscillating paths meet in an overlap space between the two adjacent portions. That is, for example, to form a wall running from right to left, one oscillating path forms the left half of the wall, another oscillating path forms the right half of the wall, and the two oscillating paths meet in an interlace region between the left half and right half of the wall. Exemplary aspects of the disclosure will be presented with reference to the individual figures, wherein like components will be identified with like numerals generally increased by 100, and with it being understood that the description of the like parts described herein carries forward unless otherwise noted.

FIG. 7 illustrates a second build plane region 452b may be irradiated along a second oscillating path 491 in a second direction D2 that is opposite the first direction D1 to consolidate an extension 471 the first wall 453a. The first wall 453a and the extension 471 can overlap 455 in an interlace region 454.

The interlace region 454 can be consolidated using a combination of the first energy beam 434a and the second energy beam 434b to help promote build quality of the three-dimensional object 114 (FIG. 1) between the first build plane region 452a and the second build plane regions 452b despite the use of multiple energy beams 134 (FIG. 1). In combination, the first build plane region 452a (irradiated by

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the first energy beam 434a), the second build plane region 452b (irradiated by the second energy beam 434b), and the interlace region 454 (irradiated by both the first energy beam 434a and the second energy beam 434b) can comprise a variety of relative configurations.

For instance, the first oscillating path 490 of the first energy beam 134a and the second oscillating path 491 of the second energy beam 434b may combine and overlap 455 to irradiate the interlace region 454. The overlap 455 refers to the portion of the build plane 128 (FIG. 1) where both the first spot 435a of the first energy beam 434a and the second spot 435b of the second energy beam 434b irradiate the same build material 118 (FIG. 1). By at least partially overlapping the first oscillating path 490 and the second oscillating path 491, the interlace region 454 may be irradiated and consolidated in a more efficient manner and mitigate potential misalignment between the first energy beam 434a and the second energy beam 434b, such as by forming a larger common melt pool in the build material 118 (FIG. 1).

The interlace region 454 can comprise a variety of relative configurations for overlapping, tapering, or otherwise joining the first wall 453a with the extension 471. For example, in some aspects of the disclosure, such as that illustrated in FIG. 7, the first oscillating path 490 and the second oscillating path 491 may overlap 455 for a defined distance. The distance and amount of overlap may depend on, for example, the configurations of the first oscillating path 490 and the second oscillating path 491, or the parameters of the first energy beam 434a and the second energy beam 434b.

FIG. 8 illustrates another embodiment of a first oscillating path 590 and a second oscillating path 591 forming a first wall 553a. In some aspects of the disclosure, such as that illustrated in FIG. 8, the amplitude A for at least one of the plurality of oscillations 592, 593 for the first oscillating path 590 and the second oscillating path 591 may vary in the interlace region 554. For example, the amplitude A may increase for the oscillations 592 of the first oscillating path 590 or the oscillations 593 of the second oscillating path 591 in the interlace region 554. As a result, the first wall 553a and the extension 571 will have a larger thickness T where they meet.

FIG. 9 illustrates a first wall 653a, wherein the first plurality of oscillations 692 of the first oscillating path 690 may taper in the interlace region 654 such that the first oscillating path 190 reduces in amplitude A as it extends towards the extension 671 until stopping or diminishing. Alternatively, or additionally, the second plurality of oscillations 693 of the second oscillating path 691 may taper in the interlace region 654 such that the second oscillating path 691 reduces in amplitude A as it extends towards the first wall 653a until stopping or diminishing.

The tapering of the first oscillating path 690 or the second oscillating path 691 may be achieved with or without overlap. For example, with reference to FIG. 9, the oscillations 692 of the first oscillating path 690 may taper towards the first side 657a or the oscillations 693 of the second oscillating path 691 may taper towards the second side 659a. In such aspects of the disclosure, the first oscillating path 690 and the second oscillating path 691 may consolidate build material 118 (FIG. 1) in the interlace region 654 without overlap.

FIG. 10 illustrates a first wall 753a, wherein the oscillations 792 of the first oscillating path 790 may taper towards the internal portion of the three-dimensional object 114 (i.e., between the first side 757a and the second side 759a) or the oscillations 793 of the second oscillating path 791 may taper towards the internal portion of the three-dimensional object

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114 (FIG. 1). In such aspects of the disclosure, the first oscillating path **790** and the second oscillating path **791** may consolidate build material **118** (FIG. 1) in the interlace region **754** with overlap **755**.

It is appreciated that while certain variations and permutations are presented herein with respect to the consolidation of the first wall **153a**, the extension **171**, and the interlace region **154**, these examples are not intended to be exclusive and other implementations may further be realized within the scope of this disclosure.

Referring additionally to FIG. 11, in some aspects of the disclosure, the three-dimensional object **114** may have a plurality of walls **853** that cross at one or more intersections **850**. While the three-dimensional object **814** presented in FIG. 11 illustrates the plurality of walls **853**, it is appreciated that any of the earlier series numbering and aspects of the disclosure (**153**, **253**, **353**, **453**, **552**, **653**, **753**) can be included. As illustrated in FIG. 11, the three-dimensional object **814** may comprise a network of walls **853** that intersect at a plurality of intersections **850**, wherein one or more of the walls **853** can be consolidated using the methods and systems disclosed herein. Such aspects of the disclosure can facilitate lattice-like support structures or represent a cross section of an interleaved thin-walled structure such as heat exchanges, flow chambers, or other design configurations suitable for additive manufacturing.

As illustrated in FIG. 11, for instance, the three-dimensional object **814** may comprise a first wall **853a**, a second wall **853b**, a third wall **853c**, a fourth wall **853d**, a fifth wall **853e**, a sixth wall **853f**, and a seventh wall **853g**. The plurality of walls may comprise distinct sets of walls **853** that are parallel such as where the first wall **853a** and the second wall **853b** form a first parallel set, the third wall **853c** and the fourth wall **853d** form a second parallel set, and the fifth wall **853e**, the sixth wall **853f**, and the seventh wall **853g** form a third parallel set. While FIG. 11 illustrates seven different walls **153** in three parallel sets, it is appreciated that the three-dimensional object **814** may comprise any number of walls **853** having any suitable configuration, with or without parallel orientations.

One or more of the walls **853** may comprise oscillations **892** such as via the first oscillating path **890**, and the second oscillating path **891**. For example, the first oscillating path **890** and the second oscillating path **891** may generally comprise the same dimensions such as the same amplitude **A**, length **L**, or distance **D** (FIG. 3). In some aspects of the disclosure, one or more of the dimensions may vary between the first oscillating path **890** and the second oscillating path **891**. Furthermore, the parameters of the energy beam used for the first oscillating path **890** and the second oscillating path **891** may be similar to one another, dissimilar to one another, or a combination thereof. For example, in some aspects of the disclosure, the amplitude **A** of the oscillations **192** may increase for one or more of, or even each of, the walls **853** at the intersections **850**. Such aspects of the disclosure may increase rigidity of the plurality of walls **853** within the three-dimensional object **814**. Alternatively, or additionally, in some aspects of the disclosure, the amplitude **A** of the oscillations **892** may decrease for one or more of, or even each of, the walls **853** at the intersections **850**.

Referring now additionally to FIG. 12, a method **300** is illustrated for additively manufacturing a three-dimensional object **114**. While reference of the method **300** is generally made hereinafter to the exemplary aspects of the disclosure illustrated in FIGS. 2-4, it is appreciated that the method **300** also applies to the exemplary aspects of the disclosure included in FIGS. 5-11. The method **300** generally com-

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prises irradiating a first build plane region **152a** using a first energy beam **134a** travelling along a first oscillating path **190** in a first direction **D1** in step **310** to consolidate a first wall **153a**, wherein a build material **118** adjacent a first side **157a** of the first wall **153a** and build material **118** adjacent a second side **159a** of the first wall **153a**, opposite the first side **157a** of the first wall **153a**, remains unconsolidated. The first oscillating path **190** can include a first plurality of oscillations **192** that define at least a first side **157a** of the first wall **153a** and a thickness **T** of the first wall **153a** in a thickness direction **DT**, perpendicular to the first direction **D1**, can be greater than a beam diameter **DB** of the first energy beam **134a**.

As discussed above, the three-dimensional object **114** may comprise a variety of relative parameters and configurations with respect to one or more walls **153** and their formation.

For example, the method **300** may include irradiating a second build plane region **152b** along a second oscillating path **210** in a second direction **D2** opposite the first direction **D1** in step **320** to consolidate the first wall **153a**. The second oscillating path **210** can include a second plurality of oscillations **212** that define a second side **159a** of the first wall **153a** opposite the first side **157a** of the first wall **153a**.

In some aspects of the disclosure, such as that illustrated in FIG. 7, the method **300** may include irradiating a second build plane region **452b** along a second oscillating path **491** in a second direction **D2** opposite the first direction **D1** in step **330** to consolidate an extension **471** of the first wall **453a** in the first direction **D1**. The second oscillating path **491** can include a second plurality of oscillations **493**. Moreover, the first plurality of oscillations **492** and the second plurality of oscillations **493** can overlap **455** in an interlace region **454**.

In some aspects of the disclosure, such as that the method **300** may include irradiating a second build plane region **152b** along a second oscillating path **210** in a second direction **D2** or third direction **D3** different than the first direction **D1** in step **340** to consolidate a second wall **853b**, wherein the first wall **853a** and the second wall **853b** intersect at an intersection **850**.

Now turning to FIG. 13, a control system **104** is illustrated. The control system **104** may be configured to perform one or more control operations associated with an additive manufacturing system **100** or an additive manufacturing machine **102**. The control operations may include one or more parts of method **300** disclosed herein, or one or more operations to otherwise consolidate the three-dimensional object **114** in accordance with the disclosure presented herein.

For example, the control operations can include irradiating a first build plane region **152a** using a first energy beam **134a** travelling along a first oscillating path **190** in a first direction **D1** to consolidate a first wall **153a**, wherein a build material **118** adjacent a first side **157a** of the first wall **153a** and build material **118** adjacent a second side **159a** of the first wall **153a**, opposite the first side **157a** of the first wall **153a**, remains unconsolidated. The first oscillating path **190** can include a first plurality of oscillations **192** that define at least a first side **157a** of the first wall **153a** and a thickness **T** of the first wall **153a** in a thickness direction **DT**, perpendicular to the first direction **D1**, can be greater than a beam diameter **DB** of the first energy beam **134a**.

As shown in FIG. 13, a control system **104** may include one or more computing devices **900**. The one or more computing devices **900** may be communicatively coupled with an additive manufacturing machine **102** or additive

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manufacturing system 100. For example, a computing device 900 may be communicatively coupled with one or more controllable components 902 of an additive manufacturing machine 102, such as one or more controllable components 902 associated with an energy beam system 132, an imaging system 158, an inertization system 160, or a positioning system 170. Additionally, or in the alternative, a computing device 900 may be communicatively coupled with a management system 106 or a user interface 108. The one or more computing devices 900 may be located locally or remotely relative to an additive manufacturing machine 102.

The one or more computing devices 900 may include one or more control modules 904 configured to cause the computing device 900 to perform one or more control operations. The one or more control modules 904 may include one or more irradiation control modules 800. The one or more control modules 904 may include control logic executable to provide control commands configured to control the one or more controllable components 902 associated with an additive manufacturing machine 102.

The one or more computing devices 900 may include one or more processors 906 and one or more memory devices 908. The one or more processors 906 may include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, or other suitable processing device. The one or more memory devices 908 may include one or more computer-readable media, including but not limited to non-transitory computer-readable media, RAM, ROM, hard drives, flash drives, or other memory devices 908. The one or more control modules 904 may be implemented at least in part by the one or more processors 906 or the one or more memory devices 908.

As used herein, the terms “processor” and “computer” and related terms, such as “processing device” and “computing device”, are not limited to just those integrated circuits referred to in the art as a computer, but broadly refers to a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific integrated circuit, and other programmable circuits, and these terms are used interchangeably herein. A memory device 908 may include, but is not limited to, a non-transitory computer-readable medium, such as a random-access memory (RAM), and computer-readable nonvolatile media, such as hard drives, flash memory, and other memory devices. Alternatively, a floppy disk, a compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), or a digital versatile disc (DVD) may also be used.

As used herein, the term “non-transitory computer-readable medium” is intended to be representative of any tangible computer-based device implemented in any method or technology for short-term and long-term storage of information, such as, computer-readable instructions, data structures, program modules and sub-modules, or other data in any device. The methods described herein may be encoded as executable instructions embodied in a tangible, non-transitory, computer readable media, including, without limitation, a storage device or a memory device. Such instructions, when executed by a processor, cause the processor to perform at least a portion of the methods described herein. Moreover, as used herein, the term “non-transitory computer-readable medium” includes all tangible, computer-readable media, including, without limitation, non-transitory computer storage devices, including, without limitation, volatile and nonvolatile media, and removable and non-removable media such as a firmware, physical and virtual storage, CD-ROMs, DVDs, and any other digital

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source such as a network or the Internet, as well as yet to be developed digital means, with the sole exception being a transitory, propagating signal.

The one or more memory devices 908 may store information accessible by the one or more processors 906, including computer-executable instructions 910 that can be executed by the one or more processors 906. The computer-executable instructions 910 may include any set of instructions which when executed by the one or more processors 906 cause the one or more processors 906 to perform operations, including calibration operations, or additive manufacturing operations. For example, the computer-executable instructions 910 may be configured to cause the one or more processors 906 to perform one or more of the steps presented in method 300 (FIG. 12) or otherwise presented in this disclosure. In some aspects of the disclosure, the computer-executable instructions 910 may be configured to cause the one or more processors 906 to irradiate the first build plane region 152a using a first energy beam 134a travelling along a first oscillating path 190 in a first direction D1 to consolidate a first wall 153a, wherein a build material 118 adjacent a first side 157a of the first wall 153a and build material 118 adjacent a second side 159a of the first wall 153a, opposite the first side 157a of the first wall 153a, remains unconsolidated. The first oscillating path 190 can include a first plurality of oscillations 192 that define at least a first side 157a of the first wall 153a and a thickness T of the first wall 153a in a thickness direction DT, perpendicular to the first direction D1, can be greater than a beam diameter DB of the first energy beam 134a.

The memory devices 908 may store a plurality of data 912 accessible by the one or more processors 906. The data 912 may be past, current, real-time, or a combination thereof. The data 912 may be stored in a data library 914. As examples, the data 912 may include data 912 associated with or generated by an additive manufacturing system 100 or an additive manufacturing machine 102, including data 912 associated with or generated by a computing device 900, an additive manufacturing machine 102, a management system 106, or a user interface 108. The data 912 may also include other data sets, parameters, outputs, information, associated with an additive manufacturing system 100 or an additive manufacturing machine 102.

The one or more computing devices 900 may also include a communication interface 916 configured to communicate with various nodes on a communication network 918 via communication lines 920 that are wired or wireless. The communication interface 916 may include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, or other suitable components. The communication interface 916 may allow the one or more computing devices 900 to communicate with various nodes associated with the additive manufacturing machine 102, the management system 106, or a user interface 108. The communication network 918 may include, for example, a local area network (LAN), a wide area network (WAN), satellite communication (SATCOM) network, very high frequency (VHF) network, a high frequency (HF) network, a wireless fidelity (Wi-Fi) network, a worldwide interoperability for microwave access (WiMAX) network, a gatelink network, or any other communication network 918 for transmitting messages to or from the computing device 900 across the communication lines 920. The communication lines 920 of communication network 918 may include a data bus or a combination of wired or wireless communication links.

The management system 106 may include a server 922 or a data warehouse 924. As an example, at least a portion of the data 912 may be stored in the data warehouse 924, and the server 922 may be configured to transmit data 912 from the data warehouse 924 to the one or more computing device 900, or to receive data 912 from the one or more computing devices 900 and to store the received data 912 in the data warehouse 924 for further purposes. The server 922 or the data warehouse 924 may be implemented as part of the one or more computing devices 900 or as part of the management system 106.

While computer devices 900 and components thereof have been disclosed herein, it is appreciated that these specific aspects of the disclosure are not intended to be limiting and alternatives may further be realized within the scope of this disclosure.

Walls 153 that are both thin and uniform can be manufactured for three-dimensional objects 114 by utilizing one or more oscillating paths 190. By utilizing a plurality of oscillations 192, the thickness of the walls 153 can be reduced while still promoting consistent and uniform consolidation of build material without excessive irradiation.

Further aspects are provided by the subject matter of the following clauses:

A method of additively manufacturing a three-dimensional object, the method comprising irradiating a first build plane region using a first energy beam defining a beam diameter, the first energy beam travelling along a first oscillating path in a first direction to consolidate a first wall defining a thickness perpendicular to the first direction, wherein a build material adjacent a first side of the first wall and the build material adjacent a second side of the first wall, opposite the first side of the first wall, remains unconsolidated, and wherein the thickness of the first wall is greater than the beam diameter.

The method of any clause herein, wherein the first oscillating path comprises a first plurality of oscillations, and wherein the thickness of the first wall is between two times and five times the beam diameter of the first energy beam.

The method of any clause herein, wherein an amplitude of the first plurality of oscillations is varied along the first oscillating path.

The method of any clause herein, wherein a power of the first energy beam is adjusted when the first plurality of oscillations is varied.

The method of any clause herein, wherein the first oscillating path comprises a first plurality of oscillations that define at least the first side of the first wall.

The method of any clause herein, further comprising irradiating a second build plane region along a second oscillating path in a second direction opposite the first direction to consolidate the first wall, wherein the first oscillating path comprises a first plurality of oscillations that define the first side of the first wall, and wherein the second oscillating path comprises a second plurality of oscillations that define the second side of the first wall.

The method of any clause herein, wherein the first energy beam irradiates the second build plane region along the second oscillating path.

The method of any clause herein, further comprising irradiating a second build plane region along a second oscillating path in a second direction opposite the first direction to consolidate an extension of the first wall in the first direction, wherein the second oscillating path comprises a second plurality of oscillations, wherein the first oscillating path comprises a first plurality of oscillations, and

wherein the first plurality of oscillations and the second plurality of oscillations overlap in an interlace region.

The method of any clause herein, wherein the first plurality of oscillations and the second plurality of oscillations taper in the interlace region.

The method of any clause herein, wherein an amplitude of at least one of the first plurality of oscillations and the second plurality of oscillations varies in the interlace region.

The method of any clause herein, wherein a second energy beam irradiates the second build plane region along the second oscillating path.

The method of any clause herein, further comprising irradiating a second build plane region along a second oscillating path comprising a second plurality of oscillations in a second direction different than the first direction to consolidate a second wall, wherein the first wall and the second wall intersect at an intersection.

The method of any clause herein wherein the first oscillating path comprises a first plurality of oscillations, and wherein an amplitude of at least one of the first plurality of oscillations and the second plurality of oscillations increases at the intersection.

The method of any clause herein, wherein a second energy beam irradiates the second build plane region along the second oscillating path.

An additive manufacturing system for additively manufacturing a three-dimensional object, the additive manufacturing system comprising, a first irradiation device configured to generate a first energy beam; a first optical assembly configured to direct the first energy beam; and a control system configured to perform one or more control operations associated with the additive manufacturing system, wherein the one or more control operations comprise irradiating a first build plane region using the first energy beam defining a beam diameter, the first energy beam travelling along a first oscillating path in a first direction to consolidate a first wall defining a thickness perpendicular to the first direction, wherein a build material adjacent a first side of the first wall and adjacent a second side of the first wall, opposite the first side of the first wall, remains unconsolidated; and wherein the thickness of the first wall is greater than the beam diameter of the first energy beam.

The additive manufacturing system of any clause herein, wherein the one or more control operations further comprise irradiating a second build plane region along a second oscillating path in a second direction opposite the first direction to consolidate the first wall, wherein the first oscillating path comprises a first plurality of oscillations, wherein the second oscillating path comprises a second plurality of oscillations that define the second side of the first wall. The additive manufacturing system of claim 15, wherein the one or more control operations further comprise irradiating a second build plane region along a second oscillating path in a second direction opposite the first direction to consolidate an extension of the first wall in the first direction, wherein the first oscillating path comprises a first plurality of oscillations, wherein the second oscillating path comprises a second plurality of oscillations, and wherein the first plurality of oscillations and the second plurality of oscillations overlap in an interlace region.

The additive manufacturing system of any clause herein, wherein the one or more control operations further comprise irradiating a second build plane region along a second oscillating path in a third direction different than the first direction to consolidate a second wall, wherein the first wall and the second wall intersect at an intersection.

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The additive manufacturing system of any clause herein, further comprising a second irradiation device configured to generate a second energy beam; and a second optical assembly configured to direct the second energy beam, wherein the second energy beam irradiates the second build plane region along the second oscillating path.

A three-dimensional object manufactured by a method comprising irradiating a first build plane region using a first energy beam defining a beam diameter, the first energy beam travelling along a first oscillating path in a first direction to consolidate a first wall defining a thickness perpendicular to the first direction, wherein a build material adjacent a first side of the first wall and the build material adjacent a second side of the first wall, opposite the first side of the first wall, remains unconsolidated, and wherein the thickness of the first wall is greater than the beam diameter.

This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

We claim:

1. A method of additively manufacturing a three-dimensional object, the method comprising:

irradiating a first build plane region using a first energy beam defining a beam diameter, the first energy beam travelling along a first oscillating path having a varying first amplitude in a first direction to consolidate a first wall defining a thickness perpendicular to the first direction, wherein a build material adjacent a first side of the first wall and the build material adjacent a second side of the first wall, opposite the first side of the first wall, remains unconsolidated, and

irradiating a second build plane region along a second oscillating path having a varying second amplitude in a second direction opposite the first direction to consolidate the first wall, wherein a portion of the first oscillating path travelling in the first direction overlaps a portion of the second oscillating path travelling in the second direction opposite the first direction,

wherein the first oscillating path includes a plurality of first oscillations that bend around and cross itself and the second oscillating path includes a plurality of second oscillations that bend around and cross itself, and wherein the plurality of first oscillations also bend around and cross the plurality of second oscillations, and

wherein the thickness of the first wall is greater than the beam diameter, wherein the plurality of first oscillations and the plurality of second oscillations overlap in an interlace region, and wherein the varying first amplitude increases in the interlace region and the varying second amplitude increases in the interlace region.

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2. The method of claim 1, wherein the thickness of the first wall is between two times and five times the beam diameter of the first energy beam.

3. The method of claim 2, wherein the varying first amplitude is different than the varying second amplitude.

4. The method of claim 3, wherein a power of the first energy beam is adjusted when the plurality of first oscillations is varied.

5. The method of claim 1, wherein the first oscillating path comprises the plurality of first oscillations that define at least the first side of the first wall.

6. The method of claim 1, wherein the first oscillating path comprises the plurality of first oscillations that define the first side of the first wall, and wherein the second oscillating path comprises the plurality of second oscillations that define the second side of the first wall, and wherein the first oscillating path is different than the second oscillating path.

7. The method of claim 6, wherein the first energy beam irradiates the second build plane region along the second oscillating path.

8. The method of claim 1, wherein irradiating the second build plane region along the second oscillating path consolidates an extension of the first wall in the first direction, wherein the first oscillating path comprises the plurality of first oscillations, wherein the second oscillating path comprises the plurality of second oscillations, and wherein the plurality of first oscillations and the plurality of second oscillations overlap in the interlace region.

9. The method of claim 8, wherein the plurality of first oscillations and the plurality of second oscillations taper in the interlace region, and wherein the plurality of first oscillations taper until stopping.

10. The method of claim 8, wherein the varying first amplitude and the varying second amplitude vary in the interlace region.

11. The method of claim 8, wherein a second energy beam irradiates the second build plane region along the second oscillating path.

12. The method of claim 1, wherein irradiating the second build plane region along the second oscillating path comprising the plurality of second oscillations in the second direction different than the first direction consolidates a second wall, wherein the first wall and the second wall intersect at an intersection.

13. The method of claim 12, wherein the varying first amplitude and the varying second amplitude increase at the intersection.

14. The method of claim 12, wherein a second energy beam irradiates the second build plane region along the second oscillating path.

15. The method of claim 1, wherein the method further comprises:

forming a network of walls that intersect at a plurality of intersections.

16. The method of claim 15, wherein the network of walls comprise the first wall, a second wall, a third wall, a fourth wall, a fifth wall, and a sixth wall, wherein the first wall and the second wall form a first parallel set, wherein the third wall and the fourth wall form a second parallel set, and wherein the fifth wall and the sixth wall form a third parallel set.

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